

AI for Bharat Hackathon

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Team Name : **TechyBitz**

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Problem Statement : Design an AI-driven solution that helps create, manage, personalize, or distribute digital content more effectively.

Briefing about the Idea:

BharatVoice AI is a multilingual voice-based assistant designed to help Indian citizens access and understand government schemes through simple voice interaction. Many government schemes are written in complex English or formal Hindi, making them difficult to understand for rural and semi-literate populations.

This solution allows users to speak in their regional language and receive simplified explanations of schemes in the same language. The system converts speech to text, processes the query using AI, retrieves relevant scheme information, simplifies it, and delivers the response through voice output.

The aim is to make government information accessible, inclusive, and easy to understand for every citizen, regardless of literacy level or language barrier.

solution

- How different is it from any of the other existing ideas?

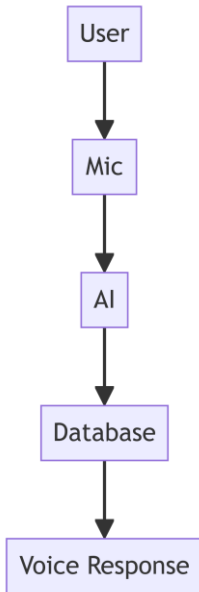
Most existing government portals are text-heavy and require users to read and interpret detailed documents. They are not voice-first and do not focus on simplification. Our solution prioritizes voice interaction and regional language accessibility.

- How will it be able to solve the problem?
 - Reduces reliance on middlemen or agents
 - Removes dependency on reading skills
 - Eliminates language barriers
 - Simplifies complex scheme language
 - Provides direct and instant responses

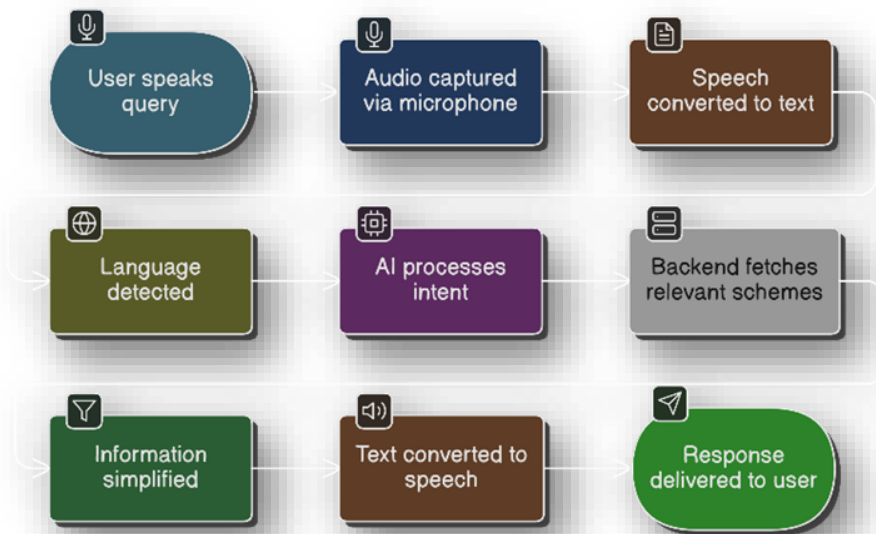
- **USP of the proposed solution**
 - Voice-first multilingual assistant
 - Automatic language detection
 - AI-based simplification of official documents
 - Designed specifically for rural and elderly users
 - Focused on accessibility and digital inclusion

List of features offered by the solution

- Query logging for system improvement
- Step-by-step guidance on application process
- Voice input in multiple Indian languages
- Automatic language detection
- Speech-to-text conversion
- AI-powered scheme recommendation
- Simplified eligibility explanation
- Text-to-speech response
- State-based filtering of schemes

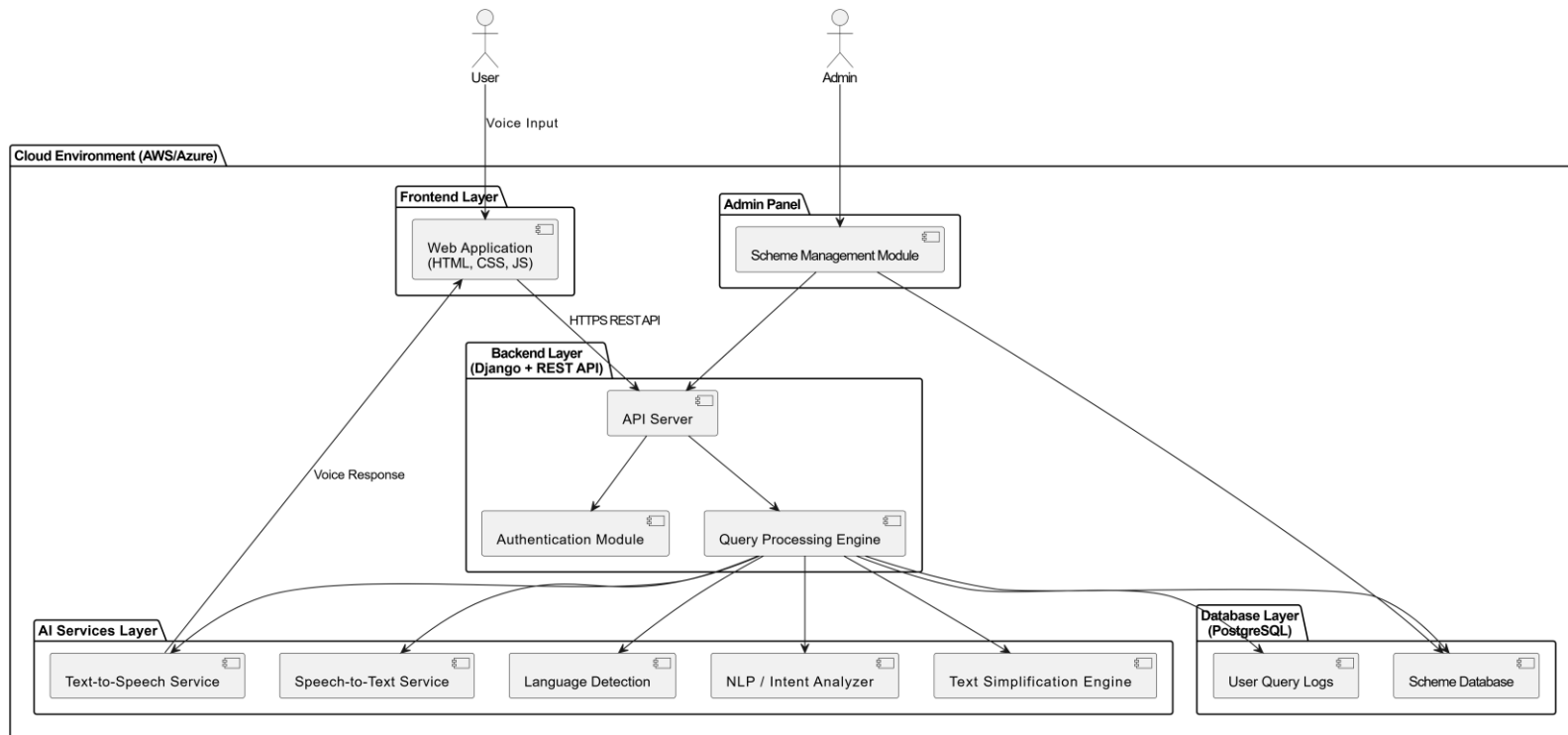


Process flow diagram



Architecture diagram of the proposed solution:

BharatVoice AI - System Architecture Diagram



Technologies to be used in the solution:

Backend:

- Python
- Django
- Django REST Framework

Frontend:

- HTML
- CSS
- JavaScript

AI Integration:

- Speech-to-Text API
- Language detection module
- Large Language Model for simplification
- Text-to-Speech API

Database:

- PostgreSQL or SQLite

Deployment:

- Cloud platform (AWS / Azure / Render)

Estimated implementation cost (optional):

Prototype Level:

- Open-source frameworks (No license cost)
- Basic cloud hosting charges
- API usage cost for speech and AI processing

Scaling Level:

- Increased AI processing cost
- Server scaling cost
- Maintenance and database scaling

Overall, initial implementation cost is low to moderate.

As per the requirements for the hackathon:

Alignment with AI for media content & digital experiences:

- Uses AI to transform complex text into simplified content
- Enhances digital interaction through voice
- Promotes multilingual accessibility

Impact:

- Improves access to government benefits
- Reduces misinformation
- Supports rural digital empowerment
- Encourages inclusive digital participation

Future Scope:

- Mobile app version
- IVR-based calling system
- WhatsApp voice assistant integration
- Integration with official government APIs

